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$$g = 10 \text{ m/s}^2$$

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**I. Write "True" if the statement is correct or "False" if the statement is wrong in the space provided on the answer sheet.**

1. Absolute pressure is the force applied parallel to a particular area.
2. A couple is a set of forces with net force but no resultant moment.
3. Different liquids have different capillary action.
4. The direction of the angular velocity is parallel to the axis of rotation.
5. Work is done against gravity when you walk up stair.
6. Torque has the same unit as work done which is Joule.
7. The sum of all torques acting on a body is zero when a body is in translational equilibrium.
8. Shear strain has no unit.

**II. Match the items under column "A" with that under column in "B".**

**Column A**

- \_\_\_1. Critical point
- \_\_\_2. Triple point
- \_\_\_3. Latent heat
- \_\_\_4. Phase diagram
- \_\_\_5. Heating curve
- \_\_\_6. Torque
- \_\_\_7. Angular momentum

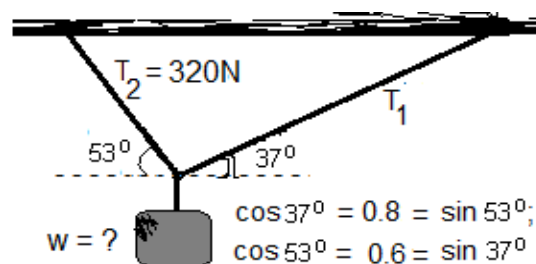
**Column B**

- A. A pressure versus temperature graph
- B. A pressure versus volume graph
- C. A temperature versus time graph
- D. Heat absorbed or released during a change of state
- E. A point in a phase diagram at which all three states of a substance are found in equilibrium
- F. A point in a phase diagram beyond which the liquid and gas phases become indistinguishable
- G.  $\vec{r} \times \vec{p}$
- H. effect of force

**III. Choose the correct answer and write the letter of your choice on the answer sheet provided.**

1. A man turns a nut through an angle of 3.2 rad by applying a torque of 40 N.m. What is the work done by the man on the nut?  
A. 68 J  
B. 128 J  
C. 75 J  
D. 136 J
2. What is the work done by  $\vec{F} = -40 \text{ N} \hat{i} - 30 \text{ N} \hat{j} + 60 \text{ N} \hat{k}$  as it pulls back on a 25 kg block that slides through  $\vec{S} = 5 \text{ m} \hat{i} + 12 \text{ m} \hat{j}$ ?  
A. 780 J  
B. - 480 J  
C. - 420 J  
D. - 560 J

3. The volume flow rate according to Poiseuille's law decrease with
- increase of pressure difference
  - increase of the radius of the pipe
  - decrease of the length of the pipe
  - increase of the coefficient of viscosity
4. A spring-mass system oscillates horizontally over a smooth surface with an amplitude of 20 cm. At what distance from the equilibrium position will its kinetic will be four-fifth of the total energy?
- $5\sqrt{3}$  cm
  - $\sqrt{6}$  cm
  - $4\sqrt{3}$  cm
  - $4\sqrt{5}$  cm
5. A body with moment of inertia  $10 \text{ kg m}^2$  is rotating at 360 rev/min. What is its angular momentum?
- 3768  $\text{kg m}^2/\text{s}$
  - 37.68  $\text{kg m}^2/\text{s}$
  - 3.768  $\text{kg m}^2/\text{s}$
  - 376.8  $\text{kg m}^2/\text{s}$
6. The magnitude of the drag force **does not** depend on
- the volume
  - the terminal velocity
  - the radius
  - the viscosity of the fluid
7. For a horizontal pipe flowing an incompressible fluid at different cross-sections according to Bernoulli's principle one of the following is unchanged
- height
  - pressure
  - velocity
  - cross-sectional area
8. One of the following is **not** a state variable.
- internal energy
  - temperature
  - pressure
  - heat capacity
9. In the figure given below, find the weight of the block suspended on the ropes if the system is in translational equilibrium.



- 400 N
  - 256 N
  - 496 N
  - 240 N
10. A diver jumps from a high board and spins in midair. Just before she reaches the water below, she extends her hands and legs and enters the water smoothly. Which one of the following physical quantities is conserved as she moves through the air?
- Angular momentum
  - Angular velocity
  - Tangential velocity
  - Speed

11. One of the following is **false** about work done. It is
  - A. the product of force and parallel distance
  - B. a scalar product
  - C. a measure of the effect of a force on a particle displacement
  - D. maximum when force and displacement are perpendicular
12. A truck of mass 2000 kg increases its velocity from rest to 30 m/s after 10 seconds. What is the average power delivered by the engine of the truck?
  - A. 90 kW
  - B. 4 W
  - C. 200 kW
  - D. 86.5 kW
13. Which one of the following is a dissipative force?
  - A. gravitational force
  - B. restoring force
  - C. frictional force
  - D. weight
14. A force  $\vec{F} = 2\vec{i} - 3\vec{j}$  N acts on a body at a point  $\vec{r} = \vec{i}$  m, what torque did it produce?
  - A.  $-4\vec{j} + 3\vec{k}$  Nm
  - B.  $-3\vec{k}$  Nm
  - C.  $-\vec{i} + \vec{k}$  Nm
  - D.  $\vec{i} + 4\vec{j} + \vec{k}$  Nm
15. If the temperature of a black body is tripled, by what factor the power radiated varies?
  - A. by 3
  - B. by 27
  - C. by 9
  - D. by 81
16. A machine part decelerates from 8 rad/s to rest at 4 rad/s<sup>2</sup> uniformly. How many times does it rotate in the mean time?
  - A. 1.58
  - B. 2.98
  - C. 1.27
  - D. 3.12
17. The rotational speed of a flywheel is 300 rev/min. Find its angular speed in SI Units.
  - A. 12.6 rad/s
  - B. 17.4 rad/s
  - C. 15.7 rad/s
  - D. 31.4 rad/s
18. What does the ratio of 'stress to strain' represent?
  - A. Stored energy
  - B. Pressure
  - C. Extension
  - D. Modulus
19. Water flows through a pipe of cross-sectional area 14 cm<sup>2</sup> at a speed of 0.4 m/s. If the flow is steady, how fast does it flow at a region where the pipe's cross-sectional area tapers to 2 cm<sup>2</sup>?
  - A. 3.4 m/s
  - B. 1.6 m/s
  - C. 2.8 m/s
  - D. 1.2 m/s
20. A certain bar conducts heat from a hot region of 340 K to a cool region of 290 K. If the bar has a cross-sectional area of 128 cm<sup>2</sup>, a length of 80 cm and a thermal conductivity of 200 W/m.K, at what rate does it conduct heat?
  - A. 140 W
  - B. 160 W
  - C. 150 W
  - D. 170 W

**IV. Fill in the blanks with an appropriate word or phrase.**

1. When you compress or stretch, energy is stored as elastic \_\_\_\_\_.
2. In an oscillation, energy is transferred between \_\_\_\_\_ and \_\_\_\_\_.
3. A point where all the mass of the body seems to be concentrated is \_\_\_\_\_.
4. The volume of liquid flowing past a given point per unit time is \_\_\_\_\_.
5. The heat energy required to raise the temperature of 1 kg of a given substance by 1 K is \_\_\_\_\_.

**V. Work out the following problems and show all the necessary steps as clearly as possible. (2 pt. each)**

1. A 80 kg box is suspended from the ceiling by a rope that has a mass 2 kg is in equilibrium.
  - a. Write an expression for the first condition of equilibrium.
  - b. What is the tension at the top of the rope?
2. A wire 3 m long with a cross-section of  $0.08 \text{ cm}^2$  hangs vertically. When a load of 200 N is applied to the wire, it stretches 3.9 mm.
  - a. Write an expression for Young's modulus
  - b. Find Young's modulus of elasticity.
3. Two forces of 60 N act as a couple. The forces act at a distance of 20 cm from the center of rotation.
  - a. Write an expression for torque.
  - b. Compute the torque produced by the couple.
4. An external force  $10\hat{i} + 4\hat{j}$  N is applied to a 2 kg mass and displaced it from  $\hat{i} + 3\hat{j}$  m to  $8\hat{i} + 4\hat{j}$  m in 2 seconds.
  - a. Write the equation used to find power
  - b. Find the power developed.
5. A ceiling fan was rotating at 0.8 rev/s. When it is turned off it slows uniformly to stop in 4 seconds.
  - a. Write the equation used to find angular displacement for the given problem.
  - b. How many revolutions did it make during this time?



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**2<sup>nd</sup> Semester Centralized Physics Final Exam for Grade 11**  
**June 2020                      Time allowed 1:30 hour                      Max. Marks: 50 %**

**Answer Sheet**

Name \_\_\_\_\_ Section \_\_\_\_\_ R. No \_\_\_\_\_

**Part I. True or False** (1 pt. each)

1. \_\_\_\_\_ 2. \_\_\_\_\_ 3. \_\_\_\_\_ 4. \_\_\_\_\_ 5. \_\_\_\_\_ 6. \_\_\_\_\_ 7. \_\_\_\_\_ 8. \_\_\_\_\_

**Part II. Match** (1 pt. each)

1. \_\_\_\_\_ 2. \_\_\_\_\_ 3. \_\_\_\_\_ 4. \_\_\_\_\_ 5. \_\_\_\_\_ 6. \_\_\_\_\_ 7. \_\_\_\_\_

**Part III: Multiple Choice** (1 pt. each)

|          |           |           |           |
|----------|-----------|-----------|-----------|
| 1. _____ | 6. _____  | 11. _____ | 16. _____ |
| 2. _____ | 7. _____  | 12. _____ | 17. _____ |
| 3. _____ | 8. _____  | 13. _____ | 18. _____ |
| 4. _____ | 9. _____  | 14. _____ | 19. _____ |
| 5. _____ | 10. _____ | 15. _____ | 20. _____ |

**Part IV: Fill in the blank** (1 pt. each)

|          |          |
|----------|----------|
| 1. _____ | 4. _____ |
| 2. _____ | 5. _____ |
| 3. _____ |          |

**Part V: Work out** (2 pt. each)

1.